

SNOWBE ONLINE Policy# LPP-04

Linux Password Procedure

Your name: Jarred Ward

Linux Password Procedure – Version 2.0

DATE: 11/23/2025

Table of Contents

PURPOSE..... 2

SCOPE..... 2

DEFINITIONS..... 2

ROLES & RESPONSIBILITIES 3

PROCEDURE..... 4

EXCEPTIONS/EXEMPTIONS 6

ENFORCEMENT 6

VERSION HISTORY TABLE 6

CITATIONS 7

Purpose

This procedure provides detailed, reproducible steps for changing, resetting, and managing passwords on Linux systems used within SnowBe's environment. It ensures that all Linux password operations follow consistent, secure practices to prevent unauthorized access. This procedure supports SnowBe's authentication requirements and helps maintain the integrity of Linux systems across development, production, and administrative environments.

Scope

This procedure applies to all SnowBe users and technical staff who access or administer Linux systems across SnowBe's development, testing, production, and cloud environments. It includes password changes, administrative resets, account locking, password expiration configuration, and secure handling of Linux credentials on systems that support standard Linux authentication tools. This procedure is intended for Linux distributions that use systemd-based logging and provide access to common administrative utilities such as `passwd`, `sudo`, `chage`, `getent`, `journalctl`, `last`, and `lastb`.

Definitions

Account Lock

A state in which a Linux user account is disabled, requiring administrative action to unlock or restore access.

Bash History

A record of previously executed shell commands that must not store or expose password information.

Password Manager

A SnowBe-approved encrypted application used to securely store Linux account credentials.

Password Vault Entry

A securely stored password record created in the password manager for a Linux system.

SSH (Secure Shell)

An encrypted method used to remotely access Linux systems through a terminal session.

Sudo

A Linux utility that allows authorized users to execute administrative commands with elevated privileges.

Username

A unique identifier assigned to a Linux account and used in authentication and administrative tasks.

Roles & Responsibilities

CISO

The CISO provides strategic oversight of SnowBe's security program and ensures that authentication and password practices across all platforms meet organizational, regulatory, and risk management requirements. The CISO establishes the policies and standards that govern how Linux passwords are created, managed, audited, and protected throughout SnowBe's environment.

Cloud Administrator

Cloud Administrators manage Linux instances hosted in SnowBe's cloud platforms and ensure password, SSH, and authentication configurations align with SnowBe security requirements. They assist with password resets and lockouts on cloud-hosted Linux servers when local administrators are not available.

Compliance Officer

Compliance Officers review Linux password practices and related evidence (such as reports, logs, and approvals) to confirm that procedures meet regulatory, contractual, and framework-based requirements. They identify gaps between the documented procedure and actual operational behavior, and drive remediation activities.

Help Desk

The Help Desk assists users who cannot log in, experience authentication issues, or need guidance after a password reset. They serve as the first point of contact when users report Linux password-related problems.

Linux Administrator

Linux Administrators perform password resets, manage account lockouts, configure password expiration, and ensure Linux authentication complies with SnowBe policy. They use sudo privileges to execute administrative password operations.

Security Operations Analyst

Security Operations Analysts investigate suspicious Linux login attempts and review authentication activity when compromise is suspected. They coordinate with administrators to ensure affected accounts are secured.

User

Any individual who authenticates into a SnowBe system, including administrators, developers, contractors, and standard users, and is responsible for securely managing their own credentials according to SnowBe's authentication requirements.

Procedure

The following materials and prerequisites are required before performing Linux password operations.

- Console or SSH access to the Linux system
- A valid Linux username
- Knowledge of SnowBe's current password complexity requirements
- Sudo privileges for administrative work
- Access to SnowBe's approved password manager

1. Change Your Own Password (All Users)

1. Log in to the Linux system using your current username and password.
2. Enter the command:
`passwd`
3. Enter your current password when prompted (characters will not display).
4. Enter a new password that meets SnowBe complexity requirements.
5. Re-enter the new password for confirmation.
6. Wait for confirmation that the password has been successfully updated.
 - Tip: Make this change during non-critical periods to avoid disrupting active sessions that may require reauthentication.

2. Reset a Forgotten Password (Administrators)

1. Log in to the Linux system using an account with sudo privileges.
2. Verify the user account exists by entering:
`getent passwd <username>`
3. Reset the user's password with:
`sudo passwd <username>`
4. Provide a temporary password that meets SnowBe requirements.
5. Instruct the user to log in and immediately change the temporary password using the following command:
`passwd`
6. Review authentication activity for the account by checking recent SSH logs:
`sudo journalctl -u sshd --since "10 minutes ago"`
7. Investigate any irregular login activity before closing the request.
 - Warning: Temporary passwords must only be delivered using SnowBe's approved secure method.

3. Lock or Unlock a Linux Account (Administrators)

1. Log in with sudo privileges.
2. To lock the account, enter:
`sudo passwd -l <username>`

3. Verify the lock by checking account status:

```
sudo passwd -S <username>
```

4. To unlock the account, enter:

```
sudo passwd -u <username>
```

5. Recheck the account status using the same command to ensure it is unlocked.

- Tip: Check for suspicious activity before unlocking any account.

4. Enforce Password Expiration Settings (Administrators)

1. Log in using sudo privileges.

2. View the user's current password aging information:

```
sudo chage -l <username>
```

3. Set the maximum password age (e.g., 90 days) by entering:

```
sudo chage -M 90 <username>
```

4. Require a user to change their password at next login by entering:

```
sudo chage -d 0 <username>
```

5. Review the updated expiration settings using the following command:

```
sudo chage -l <username>
```

- Warning: Do not modify aging parameters for privileged or service accounts without authorization.

5. Securely Store Linux Credentials (All Users)

1. Open the SnowBe-approved password manager.

2. Create a new vault entry labeled with the Linux system's hostname or purpose.

3. Store the Linux password in the encrypted vault entry, ensuring the username and system details are correct.

4. Lock or close the password vault when finished.

- Warning: Do not store passwords in shell history, scripts, shared documents, or unencrypted locations. Storing credentials in plaintext files or unencrypted locations violates SnowBe's security policy.

6. Report Suspected Linux Password or Account Compromise (All Users)

1. Immediately notify SnowBe Security Operations or the Help Desk.

2. Report any unusual authentication behavior, including unknown SSH attempts or unexpected password prompts.

3. Change your password by entering the following command:

```
passwd
```

4. Update SSH keys, MFA devices, or access methods if instructed by SnowBe Security.

5. Provide timestamps or screenshots of suspicious activity if requested by analysts.

6. Avoid using the account again until the investigation is complete.

- Tip: Use `last` and `lastb` to identify suspicious login attempts and share relevant entries with Security.

Exceptions/Exemptions

Requests for exceptions must be submitted in writing to SnowBe Online's IT Security Manager for review and approval. Each request must include the business justification, defined scope, duration, and any compensating controls that will be applied. All approved exceptions will be formally documented, reviewed periodically for continued appropriateness, and remain valid for no longer than one year unless re-evaluated and re-approved.

Enforcement

Failure to follow this policy may result in disciplinary action, beginning with a verbal warning and escalating to written notice, suspension of system access, or termination of employment. Contractors and vendors who violate SnowBe Online's security or compliance requirements may have their access revoked or contracts terminated. Intentional or severe violations, especially those involving customer data or payment systems, may lead to termination and potential legal or regulatory action under applicable laws.

Version History Table

Version #	Implementation Date	Document Owner	Approved By	Description
1.0	10/29/2025	Jarred Ward	IT Security Manager	Original Draft
2.0	11/23/2025	Jarred Ward	IT Security Manager	Updated Linux Password Procedure

Citations

Freedesktop - <https://www.freedesktop.org/software/systemd/man/latest/journalctl.html> (Used for Procedure).

Man7 - <https://man7.org/linux/man-pages/man1/passwd.1.html> (Used for Procedure).

Montana Department of Justice - <https://dojmt.gov/wp-content/uploads/Password-Policy-and-Procedures.pdf> (Used for Purpose, Scope, Enforcement).

Oracle - https://docs.oracle.com/cd/E88353_01/html/E72487/getent-8.html (Used for Procedure).

Red Hat - <https://www.redhat.com/en/blog/password-changes-chage-command> (Used for Procedure).

State of Arkansas - https://sas.arkansas.gov/wp-content/uploads/2020/04/SS-70-002_password_standard.pdf (Used for Definitions).

United States Department of Education - <https://www.ed.gov/sites/ed/files/fund/contract/about/acs/2021-password-parameters.pdf> (Used for Exceptions/Exemptions).

Weill Cornell Medicine - <https://its.weill.cornell.edu/policies/1115-password-policy> (Used for Roles & Responsibilities).